

task-1

Difference	Static Routing	Dynamic Routing	Real-World Analogy
1. Route Selection	Routes are manually configured and remain fixed unless changed by an administrator.	Routes are learned and updated automatically based on network conditions.	Google Maps: Static routing is like following a fixed route even when traffic changes; dynamic routing is like Google Maps automatically choosing another route when there is traffic.
2. Failure Handling	Does not normally find an alternative route automatically when a configured path fails.	Can automatically select an alternative available route when a path fails.	Swiggy: Static routing is like a delivery partner always using one fixed road; dynamic routing is like choosing another road if the first is blocked.
3. Management	Requires manual configuration and maintenance of routes.	Routing protocols automatically exchange information and update routes.	Google Maps: Static routing requires manually deciding the roads, while dynamic routing continuously updates directions based on current conditions.

Task-2

Q- Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

To load a pre-configured Cisco Packet Tracer network containing three routers configured with RIP and use the following commands to observe the routing information:

- show ip route
- show ip protocols

The purpose is to identify RIP-learned routes and understand their metrics.

1. Open **Cisco Packet Tracer**.
2. Open the provided pre-configured file: **first.pkt**
3. Verify that three routers are present and connected.
4. Wait for the router interfaces and RIP routing information to converge.

Step 2: Access Router R1

Click Router R1 and open the **CLI**.

Enter:

```
enable show ip  
route
```

The routing table displays directly connected routes and routes learned through RIP.

RIP routes are identified by the letter:

R

For example:

R 192.168.2.0/24 [120/1] via 10.0.12.2

R 192.168.3.0/24 [120/2] via 10.0.12.2

Here:

- R = Route learned through RIP
- 120 = Administrative Distance of RIP
- 1 or 2 = RIP metric/hop count
- via = Next-hop router

Step 3: Check RIP Configuration on R1

Enter: show ip

protocols

The command displays information about the active routing protocol.

Typical output includes:

Routing Protocol is "rip"

Sending updates every 30 seconds

Invalid after 180 seconds

Hold down 180 Flush after

240

It also displays the networks participating in RIP and information about neighboring routers.

Step 4: Repeat on Router R2 Enter:

enable show ip

route

R2 should show routes learned from R1 and R3.

Example:

R 192.168.1.0/24 [120/1] via 10.0.12.1

R 192.168.3.0/24 [120/1] via 10.0.23.3

Then enter:

show ip protocols

This confirms that RIP is running on R2.

Step 5: Repeat on Router R3 Enter:

enable show ip

route Example:

R 192.168.1.0/24 [120/2] via 10.0.23.2

R 192.168.2.0/24 [120/1] via 10.0.23.2

Then enter:

show ip protocols

This confirms that RIP is configured and exchanging routing updates.

Observation Table

Router	RIP-Learned Network	Next Hop	Metric
R1	LAN behind R2	R2	1 hop
R1	LAN behind R3	R2	2 hops
R2	LAN behind R1	R1	1 hop
R2	LAN behind R3	R3	1 hop
R3	LAN behind R2	R2	1 hop
R3	LAN behind R1	R2	2 hops

Important Output Interpretation

For a routing-table entry such as: R

192.168.3.0/24 [120/2] via 10.0.12.2

the meaning is:

Part	Meaning
R	Route was learned using RIP
192.168.3.0/24	Destination network
120	RIP administrative distance
2	RIP metric
via 10.0.12.2	Next-hop router

The **RIP metric is measured in hop counts.**

For example:

- Metric **1** = destination is one router hop away.
- Metric **2** = destination is two router hops away.
- Metric **3** = destination is three router hops away.

RIP selects the route with the **lowest hop count**.

In this experiment, RIP routing was observed on a three-router Cisco Packet Tracer network. The `show ip route` command helped identify RIP-learned routes and their metrics, while `show ip protocols` provided details about the RIP configuration.

It was observed that RIP uses **hop count as its metric**, has a default **administrative distance of 120**, and considers a maximum of **15 hops** as reachable. A metric of 16 indicates an unreachable destination.

Therefore, the experiment successfully demonstrated how routers learn remote networks dynamically using **RIP (Routing Information Protocol)**.

Task -3

Q-Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.

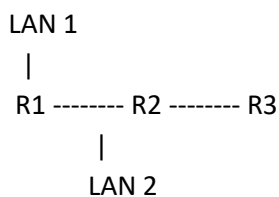
Hint: Use the `'shutdown'` command on an interface and monitor with `'show ip route'` repeatedly.

Routing Information Protocol (RIP) is a dynamic routing protocol used by routers to exchange information about available networks. RIP automatically updates routing tables when a network path becomes unavailable.

In this experiment, a network failure was simulated by shutting down one router interface in a RIP-enabled topology. The routing tables of the neighboring routers were monitored using the `show ip route` command to observe how RIP detected the failure and removed the unreachable route.

Network Topology

The experiment uses a RIP-enabled network consisting of multiple routers and connected LANs.



RIP is configured on the routers so that they learn remote networks dynamically.

Required Commands

```

show ip interface brief
show ip route
show ip route rip
show ip protocols
configure terminal
interface gigabitEthernet 0/0
shutdown
no shutdown

```

Initial Configuration Verification

Before simulating the failure, the routing table was checked on the routers.

R2# show ip route

Routes learned through RIP are identified by the letter R.

Example:

R 192.168.1.0/24 [120/1] via 10.0.0.1

Procedure

Step 1: Check the Router Interfaces

On Router R1, the interfaces were checked using:

R1# show ip interface brief

The interface connecting R1 to the rest of the RIP topology was identified.

Step 2: Shut Down the Interface

The selected interface was administratively disabled:

R1# configure terminal

R1(config)# interface gigabitEthernet 0/0 R1(config-if)# shutdown

This simulated a network-link failure.

Step 3: Monitor the Neighboring Router

Immediately after shutting down the interface, Router R2 was monitored repeatedly using:

R2# show ip route

Initially, R2 could still display the RIP route because the routing information had not yet timed out.

Step 4: Observe Route Removal

The show ip route command was repeatedly executed until the unreachable RIP route disappeared from R2's routing table.

The time between shutting down the interface and removal of the route was recorded. **7. Observation**

Event	Observation
Initial state	RIP routes were present
Interface shutdown	R1 interface became administratively down
Immediately after failure	R2 could still show the RIP route
During convergence	RIP updated its routing information
After convergence	Unreachable route was removed
Approximate convergence time	About 180–240 seconds under default RIP timers

8. RIP Convergence

RIP uses timers to determine whether a route is still reachable.

Timer	Default Value	Purpose
Update Timer	30 seconds	RIP sends routing updates
Invalid Timer	180 seconds	Route is considered invalid if updates are not received
Hold-down Timer	180 seconds	Helps prevent unstable route changes
Flush Timer	240 seconds	Route is removed from the routing table

RIP is therefore relatively slow at detecting and removing failed routes compared with many modern routing protocols.

9. Verification After Failure

The routing table was checked using:

```
R2# show ip route
```

The previously learned RIP route was no longer present after convergence.

RIP configuration and timers can also be examined using:

```
R2# show ip protocols
```

RIP-specific routes can be displayed using:

```
R2# show ip route rip
```

10. Restoring the Network

After completing the experiment, the interface was restored using:

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no shutdown
```

The routers then exchanged RIP updates again, and the previously unavailable route was eventually learned again.

The experiment successfully demonstrated RIP convergence.

When the router interface was shut down, the connected path became unavailable. The neighboring router did not immediately remove the route from its routing table. RIP required time to detect the failure through its routing-update and routing timers.

After the convergence period, the unreachable route was removed from the routing table.

Observed/expected convergence time: approximately 3–4 minutes under default RIP timing, depending on the exact failure timing and simulation environment.

Task -4

Dynamic routing in **RIP** automatically updates routes when network conditions or available paths change. Similarly, apps like **Uber or Zomato** can update delivery or travel paths when traffic, road closures, or other conditions change. RIP uses routing updates exchanged between routers, while these apps use real-time location and traffic information to suggest updated paths. In both cases, the main goal is to **automatically choose and maintain an available and suitable path** without manually changing every route.

